# Narrower Is Not Automatically Better

Unit 3 · Topic 3.3 · TODAY'S PROBLEM

A library system has 20,000 adult cardholders and wants to estimate $p$, the proportion who used a digital audiobook last month.

**Plan A:** A computer takes an SRS of 200 cardholder IDs. All answer; 116 say yes.

**Plan B:** A newsletter link lets cardholders choose to respond once. Of 800 respondents, 520 say yes.

**Eli:** "Use Plan B. Its larger sample gives the narrower interval."

**Dana:** "Use Plan A. Entry into the sample matters more than the narrowest output."

**Plan counts**

| Plan | N | n | Yes | No |
|---|---|---|---|---|
| A | 20,000 | 200 | 116 | 84 |
| B | 20,000 | 800 | 520 | 280 |

**FIRST TAKE (5 min, on your sheet)**

Which plan, if either, should estimate all adult cardholders? Cite one collection detail.

DOK 1 **(a)** Define the population parameter $p$ in context and name the confidence-interval procedure that would be appropriate when its conditions are met.

DOK 2 **(b)** For each plan, compute $\hat{p}$, check the random, 10 percent, and large-count conditions, and calculate the 95 percent formula interval. Mark any nondefensible population interval.

DOK 3 **(c)** Decide which plan can support a 95 percent interval for all cardholders. Cite recruitment and a numerical condition, compare interval widths, and explain the reader consequence of reporting Plan B's output.

**Video + follow-along:** `u6_lesson1-2_live.html` (scan the code)

**Finish (a)–(c) after the video. Turn in your sheet.**